

Procedure for the preparation of 1 grain of KNSB Composite Sugar Propellant

March 13, 2026

1 Apparatus and Reagents

Apparatus

Fine potassium nitrate
Sorbitol solution

Reagents

Casting mold
Liner
Core rod
Stopwatch
Skillet pan
Induction stove
Electrical balance
Silicon spatula
Grinder
Grease
Plastic container

2 Method

1. Place the skillet pan on the induction stove.
2. Weigh 690.87 g of sorbitol into the skillet pan and set the temperature on the stove to 150°C.



Figure 1: Heating sorbitol syrup in a skillet

3. As the sorbitol solution comes to a boil, weigh 944.84 g of fine potassium nitrate into the plastic container. Check whether the potassium nitrate is a fine powder. If not, use the grinder.



Figure 2: Weighing potassium nitrate



Figure 3: Grinding potassium nitrate to fine powder



Figure 4: Stirring the mixture after the potassium nitrate has been added

4. Assemble the casting mold as shown, with the exception of the core rod and top plate.
5. After 30 minutes, reduce the temperature from 150°C to 120°C and begin stirring the solution while checking for a colour change to caramel.
6. After 5 minutes (or 1-2 minutes longer), reduce the temperature to 90°C and begin adding small amounts of fine potassium nitrate powder, whilst stirring. After 1 minute, increase the temperature to 120°C
7. After all the potassium nitrate powder has been added and 20 minutes have elapsed, reduce the temperature to 60°C while stirring.
8. Transfer the contents of the skillet into the casting mold and close it as shown.



Figure 5: Grain set up to cure under pressure

9. Set the now casted grain aside for 5 days to a week.



Figure 6: Cured grain

3 Safety Precautions

- Wear PPE (Personal Protective Equipment) while preparing the grain.

- Avoid areas with open flame while preparing the grain and keep a fire extinguisher nearby.
- Work in a well lit and ventilated area.
- Avoid distractions like cell phones or listening to music while cooking.
- Grind potassium nitrate carefully at a low setting to avoid sparking
- Use pure ingredients with minimal contamination to avoid spontaneous detonation